

6(a)	work done / charge or energy (transferred from electrical to other forms) / charge	B1
6(b)	for $V < 0.25 \text{ V}$ resistance is infinite/very high (as current is zero)	B1
	for $V > 0.25 \text{ V}$ resistance decreases (as V increases)	B1
6(c)(i)	$R = V / I$	C1
	$= 0.75 / (15 \times 10^{-3})$	C1
	$= 50 \Omega$	A1

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Question	Answer	Marks
6(c)(ii)	1. $V_Y = 15 \times 10^{-3} \times 60$ (= 0.90 V)	C1
	$V_X = 2.0 - 0.90 - 0.75$ (= 0.35 V)	C1
	$R_X = 0.35 / (15 \times 10^{-3})$ $= 23 \Omega$	A1
	or	
	total $R = 60 + 50 + R_X$	(C1)
	$60 + 50 + R_X = 2.0 / (15 \times 10^{-3})$	(C1)
	$R_X = 23 \Omega$	(A1)
	2. $P = VI$ or $P = EI$ or $P = I^2R$ or $P = V^2 / R$	C1
	ratio = $\frac{(15 \times 10^{-3})^2 \times 60}{2.0 \times 15 \times 10^{-3}}$ or $\frac{0.90 \times 15 \times 10^{-3}}{2.0 \times 15 \times 10^{-3}}$ or $\frac{(0.90^2 / 60)}{2.0 \times 15 \times 10^{-3}}$ $= 0.45$	A1